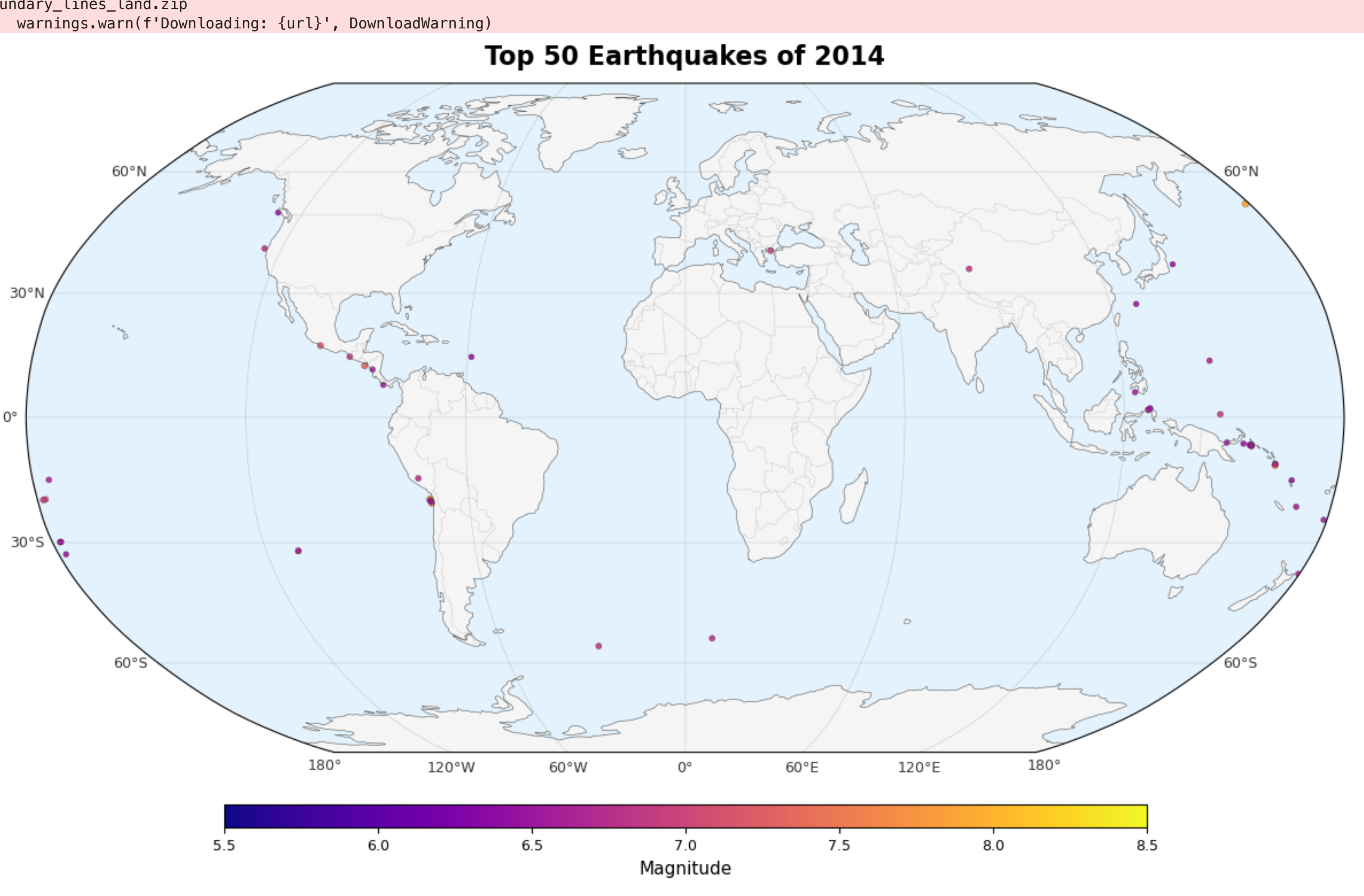
In [5]:

```python
import pandas as pd
import matplotlib.pyplot as plt
import cartopy.crs as ccrs
import cartopy.feature as cfeature
import numpy as np

def plot_top50_2014():
    try:
        df = pd.read_csv('usgs_earthquakes.csv')
    except FileNotFoundError:
        print("错误：找不到 usgs_earthquakes.csv 文件，请确保文件在当前目录下。")
        return

    df = df.sort_values('mag', ascending=False).head(50)

    lats = df['latitude'].values
    lons = df['longitude'].values
    mags = df['mag'].values

    fig = plt.figure(figsize=(12, 8))
    ax = plt.axes(projection=ccrs.Robinson())
    ax.set_global()

    ax.add_feature(cfeature.LAND, facecolor='#F5F5F5', linewidth=0)
    ax.add_feature(cfeature.OCEAN, facecolor='#E3F2FD')
    ax.add_feature(cfeature.COASTLINE, edgecolor='#555555', linewidth=0.4)
    ax.add_feature(cfeature.BORDERS, linestyle=':', edgecolor='#777777', linewidth=0.3)

    gl = ax.gridlines(draw_labels=True, linewidth=0.3,
                      color='#999999', alpha=0.6, linestyle='-')
    gl.top_labels = gl.right_labels = False
    gl.xlabel_style = {'size': 9, 'color': '#333333'}
    gl.ylabel_style = {'size': 9, 'color': '#333333'}

    scale = 28 / (8.2 ** 2)
    sizes = (mags ** 2) * scale * 72 / fig.dpi  # 转 points

    cmap = plt.cm.get_cmap('plasma')
    norm = plt.Normalize(vmin=5.5, vmax=8.5)
    colors = cmap(norm(mags))

    scatter = ax.scatter(lons, lats, s=sizes, c=colors,
                         transform=ccrs.PlateCarree(),
                         edgecolor='black', linewidth=0.15, alpha=0.9)

    plt.title('Top 50 Earthquakes of 2014', fontsize=16, pad=12, weight='bold')

    cbar = plt.colorbar(plt.cm.ScalarMappable(norm=norm, cmap=cmap),
                        ax=ax, orientation='horizontal',
                        pad=0.06, shrink=0.7, aspect=40)
    cbar.set_label('Magnitude', fontsize=11)
    cbar.set_ticks([5.5, 6, 6.5, 7, 7.5, 8, 8.5])
    cbar.ax.tick_params(labelsize=9)

    plt.tight_layout()
    plt.show()

if __name__ == "__main__":
    plot_top50_2014()
```

```
/var/folders/x8/8y9n4rkd2_g5631mlytyr71r0000gn/T/ipykernel_62942/470127885.py:45: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap = plt.cm.get_cmap('plasma')
/opt/anaconda3/lib/python3.13/site-packages/cartopy/io/__init__.py:242: DownloadWarning: Downloading: https://naturalearth.s3.amazonaws.com/110m_cultural/ne_110m_admin_0_boundary_lines_land.zip
  warnings.warn(f'Downloading: {url}', DownloadWarning)
```



In [ ]:

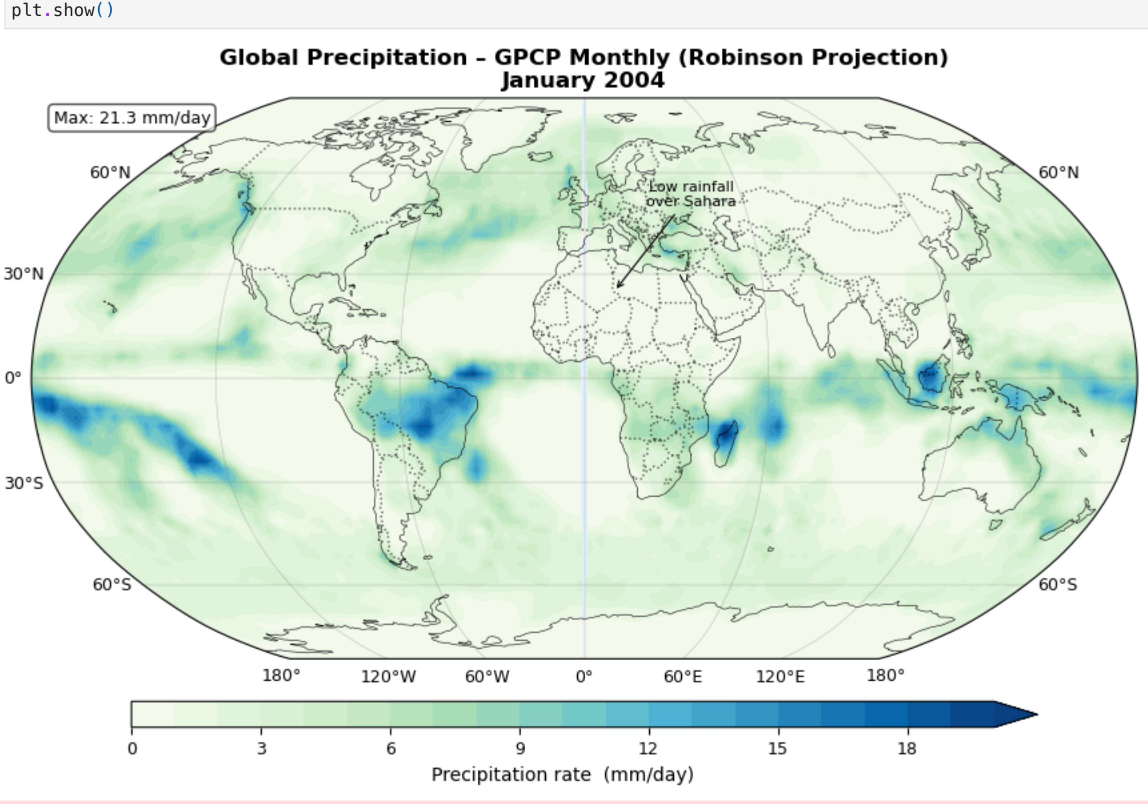
In [7]:

```python
import xarray as xr

# 打开你的 netCDF 文件
ds = xr.open_dataset('precip.mon.mean.nc')

# 打印数据集信息
print(ds.info())
```

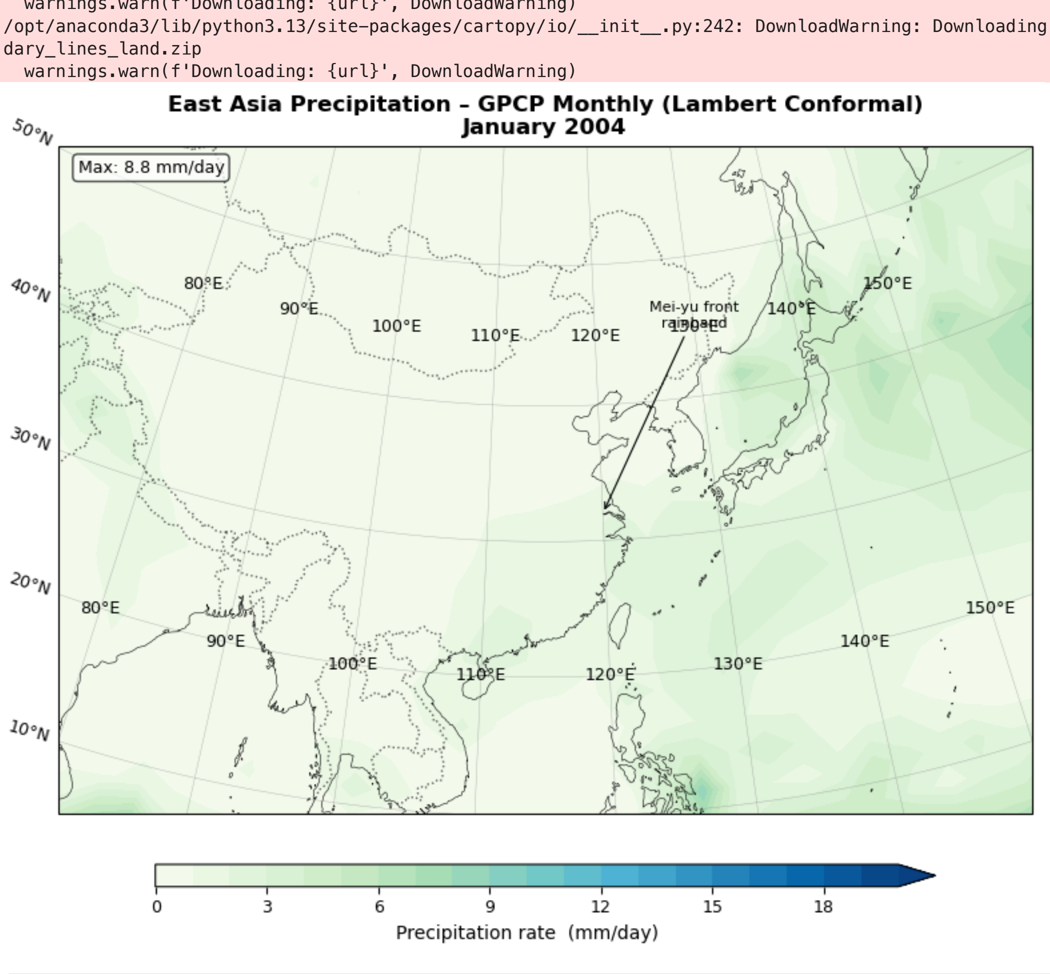
```
xarray.Dataset {
dimensions:
	time = 561 ;
	nv = 2 ;
	lat = 72 ;
	lon = 144 ;

variables:
	datetime64[ns] time_bnds(time, nv) ;
		time_bnds:comment = time bounds for each time value ;
	float32 lat_bnds(lat, nv) ;
		lat_bnds:units = degrees_north ;
		lat_bnds:comment = latitude values at the north and south bounds of each pixel. ;
	float32 lon_bnds(lon, nv) ;
		lon_bnds:units = degrees_east ;
		lon_bnds:comment = longitude values at the west and east bounds of each pixel. ;
	float32 precip(time, lat, lon) ;
		precip:long_name = Average Monthly Rate of Precipitation ;
		precip:valid_range = [  0. 100.] ;
		precip:units = mm/day ;
		precip:precision = 32767 ;
		precip:var_desc = Precipitation ;
		precip:dataset = GPCP Version 2.3 Combined Precipitation Dataset ;
		precip:level_desc = Surface ;
		precip:statistic = Mean ;
		precip:parent_stat = Mean ;
		precip:actual_range = [0.000000e+00 1.365714e+13] ;
	float32 lat(lat) ;
		lat:units = degrees_north ;
		lat:actual_range = [ 88.75 -88.75] ;
		lat:long_name = Latitude ;
		lat:standard_name = latitude ;
		lat:axis = Y ;
	float32 lon(lon) ;
		lon:units = degrees_east ;
		lon:long_name = Longitude ;
		lon:actual_range = [  1.25 358.75] ;
		lon:standard_name = longitude ;
		lon:axis = X ;
	datetime64[ns] time(time) ;
		time:long_name = Time ;
		time:delta_t = 0000-01-00 00:00:00 ;
		time:avg_period = 0000-01-00 00:00:00 ;
		time:standard_name = time ;
		time:axis = T ;
		time:actual_range = [65378. 82423.] ;

// global attributes:
	:Conventions = CF-1.0 ;
	:curator = Dr. Jian-Jian Wang
ESSIC, University of Maryland College Park
College Park, MD  20742  USA
Phone: +1 301-405-4887 ;
	:citation = Adler, R.F., G.J. Huffman, A. Chang, R. Ferraro, P. Xie, J. Janowiak, B.
Rudolf, U. Schneider, S. Curtis, D. Bolvin, A. Gruber, J. Susskind, P.
Arkin, 2003: The Version 2 Global Precipitation Climatology Project
(GPCP) Monthly Precipitation Analysis (1979 - Present). J. Hydrometeor.,
4(6), 1147-1167. ;
	:title = GPCP Version 2.3 Combined Precipitation Dataset (Final) ;
	:platform = NOAA POES (Polar Orbiting Environmental Satellites) ;
	:source_obs = CDR RSS SSMI/SSMIS Tbs over ocean
CDR SSMI/SSMIS rainrates over land (Ferraro)
Geo-IR (Xie) calibrated by SSMI/SSMIS rainrates for sampling
TOVS/AIRS empirical precipitation estimates at higher latitudes
(ocean and land)
GPCC gauge analysis to bias correct satellite estimates over land and
merge with satellite based on sampling
OLR Precipitation Index (OPI) (Xie) used for period before 1988 ;
	:documentation = http://www.esrl.noaa.gov/psd/data/gridded/data.gpcp.html ;
	:version = V2.3 ;
	:Acknowledgement =
 ;
	:contributor_name = Robert Adler    University of Maryland
George Huffman  NASA Goddard Space Flight Center
David Bolvin    NASA Goddard Space Flight Center/SSAI
Eric Nelkin     NASA Goddard Space Flight Center/SSAI
Udo Schneider   GPCC, Deutscher Wetterdienst
Andreas Becker  GPCC, Deutscher Wetterdienst
Long Chiu       George Mason University
Mathew Sapiano  University of Maryland
Pingping Xie    Climate Prediction Center, NWS, NOAA
Ralph Ferraro   NESDIS, NOAA
Jian-Jian Wang  University of Maryland
Guojun Gu       University of Maryland ;
	:dataset_title = Global Precipitation Climatology Project (GPCP) Monthly Analysis Product ;
	:description = https://data.nodc.noaa.gov/cgi-bin/iso?id=gov.noaa.ncdc:C00970 ;
	:source = https://www.ncei.noaa.gov/data/global-precipitation-climatology-project-gpcp-monthly/access/ ;
	:source_documentation = https://www.ncdc.noaa.gov/cdr/atmospheric/precipitation-gpcp-monthly ;
	:NCO = 4.6.9 ;
	:history = Generated at NOAA/ESRL PSD Sep 9 2016 based on data from source
and stored in single netCDF4 file ;
	:References = http://www.psl.noaa.gov/data/gridded/data.gpcp.html ;
	:data_comment = Interim data covers 2025/08 through latest. ;
}None
```

In [9]:

```python
import xarray as xr
import cartopy.crs as ccrs
import cartopy.feature as cfeat
import matplotlib.pyplot as plt
import numpy as np
import cartopy.mpl.ticker as cticker

ds   = xr.open_dataset('precip.mon.mean.nc')
t_idx = 300
pr    = ds['precip'].isel(time=t_idx)
lon   = pr.lon
lat   = pr.lat

#2.1 全球地图
proj_glob = ccrs.Robinson(central_longitude=0)
fig1 = plt.figure(figsize=(10, 6))
ax1  = fig1.add_subplot(111, projection=proj_glob)
ax1.set_global()

ax1.add_feature(cfeat.LAND, facecolor='#F5F5F5', linewidth=0)
ax1.add_feature(cfeat.OCEAN, facecolor='#E3F2FD')
ax1.add_feature(cfeat.COASTLINE, linewidth=0.4)
ax1.add_feature(cfeat.BORDERS, linestyle=':', alpha=0.6)

gl1 = ax1.gridlines(draw_labels=True, linewidth=0.3,
                    xlocs=np.arange(-180,181,60),
                    ylocs=np.arange(-60,61,30),
                    color='gray', alpha=0.6, linestyle='-')
gl1.top_labels   = False
gl1.right_labels = False
gl1.xlabel_style = {'size': 9}
gl1.ylabel_style = {'size': 9}

levels = np.linspace(0, 20, 21)
cmap   = plt.cm.GnBu
cf1 = ax1.contourf(lon, lat, pr, levels=levels, cmap=cmap,
                   transform=ccrs.PlateCarree(), extend='max')

cbar1 = fig1.colorbar(cf1, ax=ax1, orientation='horizontal',
                      pad=0.06, shrink=0.7, aspect=35)
cbar1.set_label('Precipitation rate  (mm/day)', fontsize=10)
cbar1.ax.tick_params(labelsize=9)

ax1.set_title('Global Precipitation – GPCP Monthly (Robinson Projection)\n'
              f'{pr.time.dt.strftime("%B %Y").item()}',
              fontsize=12, weight='bold')

ax1.text(0.02, 0.98, f'Max: {pr.max().item():.1f} mm/day',
         transform=ax1.transAxes, ha='left', va='top',
         bbox=dict(boxstyle='round', facecolor='w', alpha=0.7),
         fontsize=9)
ax1.annotate('Low rainfall\nover Sahara', xy=(10, 25),
             xycoords=ccrs.PlateCarree()._as_mpl_transform(ax1),
             xytext=(40, 50), textcoords=ccrs.PlateCarree()._as_mpl_transform(ax1),
             arrowprops=dict(arrowstyle='->', color='k', lw=0.8),
             fontsize=8, ha='center')

plt.tight_layout()
plt.show()

#2.2 区域地图（东亚）
proj_reg = ccrs.LambertConformal(central_longitude=115,
                                 central_latitude=30,
                                 standard_parallels=(20, 40))
fig2 = plt.figure(figsize=(8, 7))
ax2  = fig2.add_subplot(111, projection=proj_reg)
ax2.set_extent([80, 150, 10, 55], crs=ccrs.PlateCarree())

ax2.add_feature(cfeat.LAND.with_scale('50m'), facecolor='#F5F5F5', linewidth=0.3)
ax2.add_feature(cfeat.OCEAN.with_scale('50m'), facecolor='#E3F2FD')
ax2.add_feature(cfeat.COASTLINE.with_scale('50m'), linewidth=0.4)
ax2.add_feature(cfeat.BORDERS.with_scale('50m'), linestyle=':', alpha=0.6)

gl2 = ax2.gridlines(draw_labels=True, linewidth=0.3,
                    xlocs=np.arange(80, 151, 10),
                    ylocs=np.arange(10, 56, 10),
                    color='gray', alpha=0.6, linestyle='-')
gl2.top_labels   = False
gl2.right_labels = False
gl2.xlabel_style = {'size': 9}
gl2.ylabel_style = {'size': 9}

ax2.set_xlabel('Longitude', fontsize=10)
ax2.set_ylabel('Latitude', fontsize=10)

cf2 = ax2.contourf(lon, lat, pr, levels=levels, cmap=cmap,
                   transform=ccrs.PlateCarree(), extend='max')

cbar2 = fig2.colorbar(cf2, ax=ax2, orientation='horizontal',
                      pad=0.06, shrink=0.8, aspect=35)
cbar2.set_label('Precipitation rate  (mm/day)', fontsize=10)
cbar2.ax.tick_params(labelsize=9)

ax2.set_title('East Asia Precipitation – GPCP Monthly (Lambert Conformal)\n'
              f'{pr.time.dt.strftime("%B %Y").item()}',
              fontsize=12, weight='bold')

ax2.text(0.02, 0.98, f'Max: {pr.sel(lat=slice(10, 55), lon=slice(80, 150)).max().item():.1f} mm/day',
         transform=ax2.transAxes, ha='left', va='top',
         bbox=dict(boxstyle='round', facecolor='w', alpha=0.7),
         fontsize=9)
ax2.annotate('Mei-yu front\nrainband', xy=(120, 32),
             xycoords=ccrs.PlateCarree()._as_mpl_transform(ax2),
             xytext=(130, 45), textcoords=ccrs.PlateCarree()._as_mpl_transform(ax2),
             arrowprops=dict(arrowstyle='->', color='k', lw=0.8),
             fontsize=8, ha='center')

plt.tight_layout()
plt.show()
```




```
/opt/anaconda3/lib/python3.13/site-packages/cartopy/io/__init__.py:242: DownloadWarning: Downloading: https://naturalearth.s3.amazonaws.com/50m_physical/ne_50m_land.zip
  warnings.warn(f'Downloading: {url}', DownloadWarning)
/opt/anaconda3/lib/python3.13/site-packages/cartopy/io/__init__.py:242: DownloadWarning: Downloading: https://naturalearth.s3.amazonaws.com/50m_physical/ne_50m_ocean.zip
  warnings.warn(f'Downloading: {url}', DownloadWarning)
/opt/anaconda3/lib/python3.13/site-packages/cartopy/io/__init__.py:242: DownloadWarning: Downloading: https://naturalearth.s3.amazonaws.com/50m_physical/ne_50m_coastline.zip
  warnings.warn(f'Downloading: {url}', DownloadWarning)
/opt/anaconda3/lib/python3.13/site-packages/cartopy/io/__init__.py:242: DownloadWarning: Downloading: https://naturalearth.s3.amazonaws.com/50m_cultural/ne_50m_admin_0_boundary_lines_land.zip
  warnings.warn(f'Downloading: {url}', DownloadWarning)
```



In [ ]: